# NM2 - cv4

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## 1 Lorenzovy rovnice

$$\dot{u}_1 = \sigma(u_2 - u_1)$$
$$\dot{u}_2 = \rho u_1 - u_2 + u_1 u_3$$
$$\dot{u}_3 = -\beta u_3 + u_1 u_2$$

- Interval:$[0, 100]$
- u(1) = -1.3591409142295225
- Integrační krok: $10^{-2}$
- Eulerova metoda

### 1.1 Změna v parametru $\beta$

Table 1: Analýza v parametru $\beta$

| $\sigma$ | $\rho$ | $\beta$ | $u_1(0)$ | $u_2(0)$ | $u_3(0)$ |
|---|---|---|---|---|---|
| 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| 1.0 | 1.0 | 6.0 | 1.0 | 1.0 | 1.0 |
| 1.0 | 1.0 | 11.0 | 1.0 | 1.0 | 1.0 |
| 1.0 | 1.0 | 16.0 | 1.0 | 1.0 | 1.0 |
| 1.0 | 1.0 | 21.0 | 1.0 | 1.0 | 1.0 |

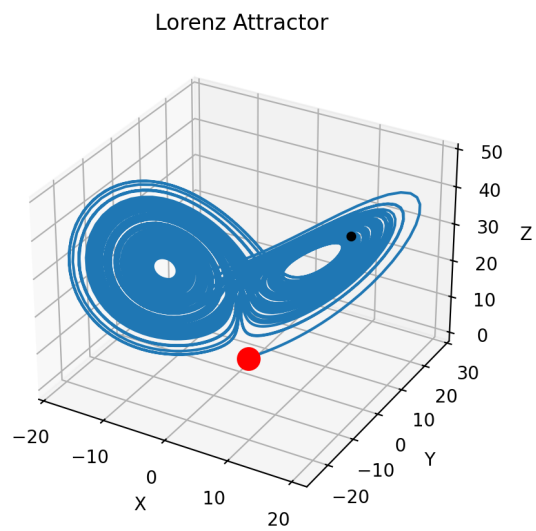


Figure 1: $\beta = 1$

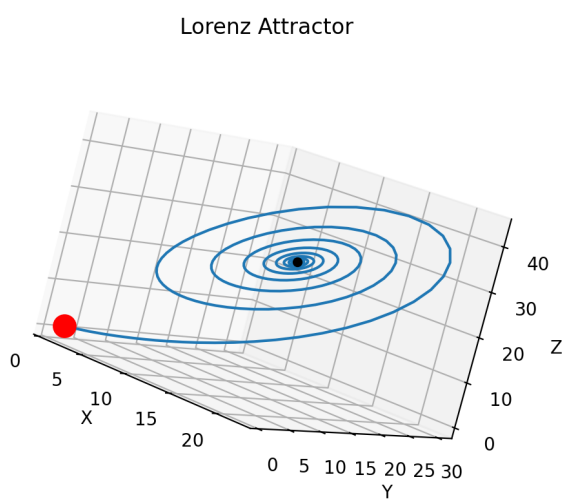


Figure 2: $\beta = 6$

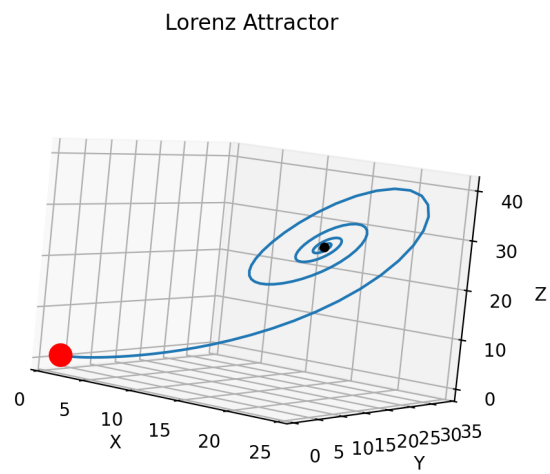


Figure 3: $\beta = 11$

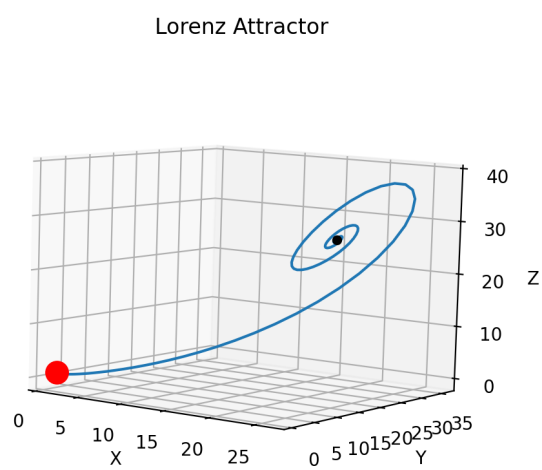


Figure 4: $\beta = 16$

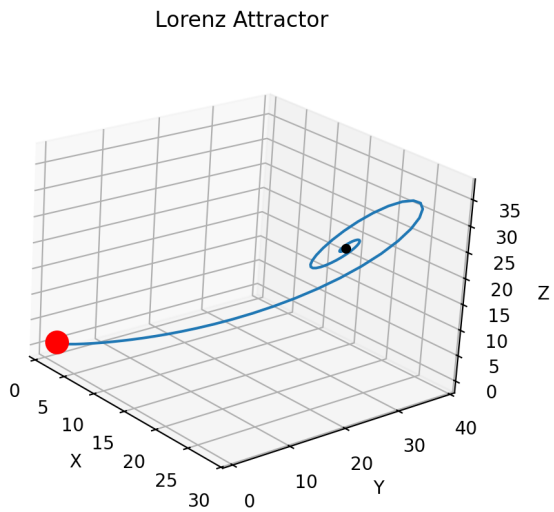


Figure 5: $\beta = 21$

## 1.2 Změna v počátečních podmínkách

Table 2: Analýza v počátečních podmínkách

| $\sigma$ | $\rho$ | $\beta$ | $u_1(0)$ | $u_2(0)$ | $u_3(0)$ |
|---|---|---|---|---|---|
| 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| 1.0 | 1.0 | 1.0 | 31.0 | 31.0 | 31.0 |
| 1.0 | 1.0 | 1.0 | 61.0 | 61.0 | 61.0 |
| 1.0 | 1.0 | 1.0 | 91.0 | 91.0 | 91.0 |
| 1.0 | 1.0 | 1.0 | 121.0 | 121.0 | 121.0 |

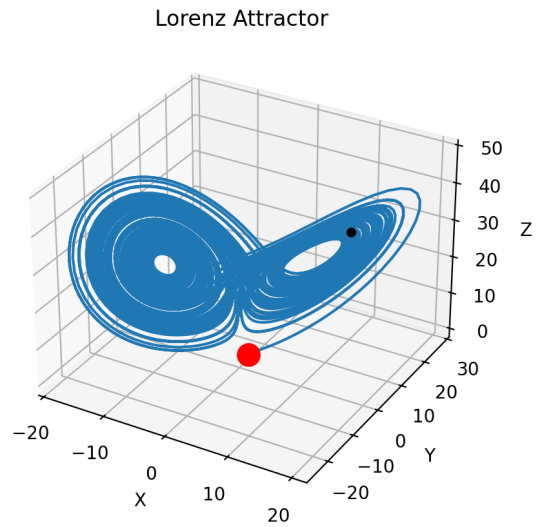


Figure 6: $u_1(0) = 1, u_2(0) = 1, u_3(0) = 1$

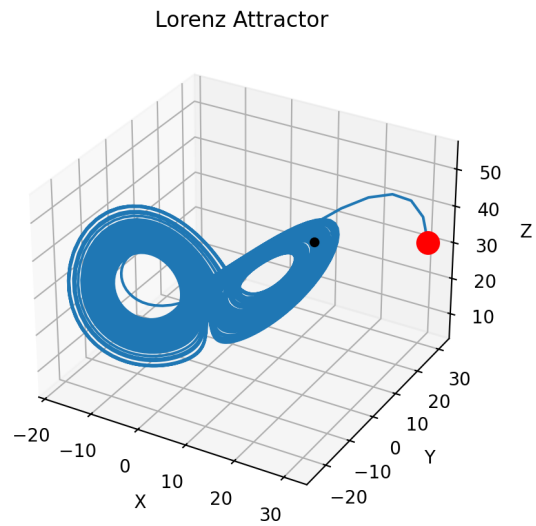


Figure 7: $u_1(0) = 31, u_2(0) = 31, u_3(0) = 31$

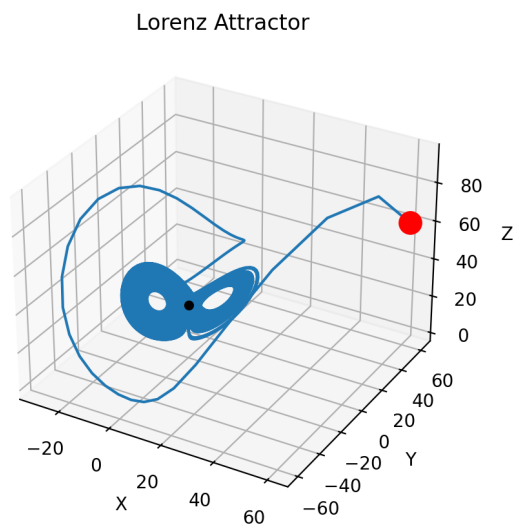


Figure 8: $u_1(0) = 61, u_2(0) = 61, u_3(0) = 61$

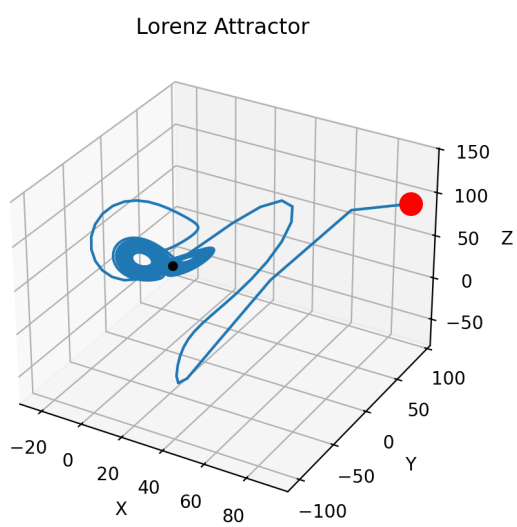


Figure 9: $u_1(0) = 91, u_2(0) = 91, u_3(0) = 91$

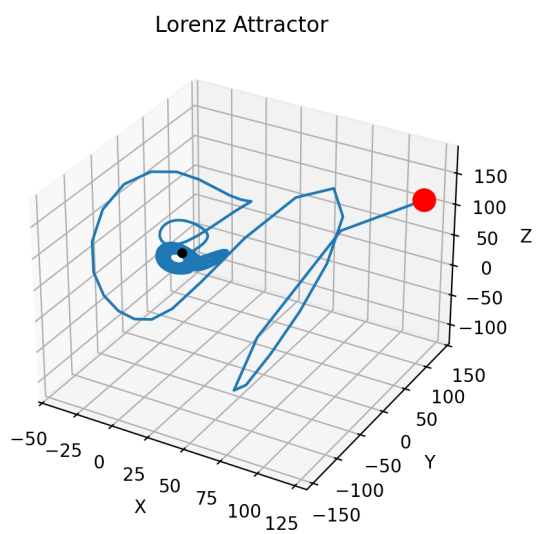


Figure 10: $u_1(0) = 121, u_2(0) = 121, u_3(0) = 121$